# Project Summary Report

Executive Summary

Headline:

AlphaEarth embeddings predict Sentinel-1 backscatter strongly for VV and VH, more moderately for polarization-derived targets, and show a weaker but still measurable cross-modal similarity signal.

Top results:

- S1_VV: best result is GBDT (LightGBM) / Embeddings + region + Dynamic World with R2=0.902.

  Ridge / Embeddings only: R2=0.868, RMSE=1.456, MAE=0.938

  Ridge / Embeddings + region + Dynamic World: R2=0.870, RMSE=1.445, MAE=0.934

  GBDT (LightGBM) / Embeddings + region + Dynamic World: R2=0.902, RMSE=1.260, MAE=0.824

  GBDT (LightGBM) / Embeddings only: R2=0.900, RMSE=1.272, MAE=0.834

- S1_VH: best result is GBDT (LightGBM) / Embeddings + region + Dynamic World with R2=0.943.

  Ridge / Embeddings only: R2=0.929, RMSE=1.317, MAE=0.945

  Ridge / Embeddings + region + Dynamic World: R2=0.929, RMSE=1.315, MAE=0.944

  GBDT (LightGBM) / Embeddings + region + Dynamic World: R2=0.943, RMSE=1.175, MAE=0.858

  GBDT (LightGBM) / Embeddings only: R2=0.941, RMSE=1.194, MAE=0.861

Polarization-difference result:

- S1_VV_div_VH: best result is Ridge direct / Embeddings + region + Dynamic World with R2=0.708.

# Project Summary Report (cont.)

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Ridge direct / Embeddings only: R2=0.708, RMSE=1.137, MAE=0.801

Structural Ridge baseline / Embeddings only: R2=0.708, RMSE=1.137, MAE=0.801

GBDT (LightGBM) / Embeddings only: R2=0.707, RMSE=1.139, MAE=0.796

Ridge direct / Embeddings + region + Dynamic World: R2=0.708, RMSE=1.137,
MAE=0.801


Structural Ridge baseline / Embeddings + region + Dynamic World: R2=0.708,
RMSE=1.137, MAE=0.801


GBDT (LightGBM) / Embeddings + region + Dynamic World: R2=0.699, RMSE=1.154,
MAE=0.798
```

- Derived linear VV/VH benchmark: LightGBM direct ratio (Embeddings only) is strongest at R2=0.584.

Failure-mode readout:

Highest-error S1_VV land-use classes by MAE:

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built: MAE=1.227, bias=-0.333

water: MAE=1.011, bias=0.020

shrub_and_scrub: MAE=0.902, bias=-0.054
```

Cross-modal similarity readout:

- At k=10, mean embedding/SAR neighbor overlap is 0.032 versus a random baseline near 0.003.

- Overall embedding-to-SAR distance correlation is modest (Pearson r=0.195), but within-region correlation peaks at 0.689 in california_coast.

Executive takeaway:

# Project Summary Report (cont.)

- The embeddings carry clear SAR-relevant signal. The strongest wins are on VV and VH, polarization structure is recoverable but weaker, and cross-modal similarity is real but much more convincing within region than across the pooled dataset.

# Plan Alignment

Working Plan Alignment

Source plan: alphaearth-to-sar-cross-model-prediction-eda-working-plan.md

Original planned workflow:

- Phase 1: Data acquisition and preparation

- Phase 2: Regression modeling

- Phase 3: Failure-mode analysis

- Phase 4: Cross-modal similarity

What was actually completed:

- A full-dataset workflow on 2,880 balanced samples instead of the earlier small-subset concept.

- Phase 2 used LightGBM as the primary nonlinear model, with added Ridge-based follow-up baselines for polarization difference and derived ratio analysis.

- Phase 3 and Phase 4 were completed with reportable outputs and summary figures.

- Umbra SAR remains a planned extension rather than a completed dataset in the current pipeline.

Interpretation:

- The project should be described as an updated implementation of the working plan, not a verbatim execution of every proposed modeling choice.

# Methods and Data

Methods and Data

Dataset:

- 2,880 co-located samples across four regions and nine Dynamic World labels.

- Targets: S1_VV, S1_VH, and S1_VV_div_VH where the stored polarization target is the dB-space difference VV - VH.

- Feature sets: 64-dim AlphaEarth embeddings alone, or embeddings plus region and Dynamic World context.

Evaluation design:

- 70/30 held-out split stratified by region x land-use label.

- Repeated grouped cross-validation on the training split using spatial blocks inside each region.

- Metrics: R2, RMSE, MAE, Pearson correlation.

Caveat from the working plan context:

- The holdout is stratified random rather than geographically disjoint, so performance should be interpreted as within-region generalization rather than strict transfer to unseen geographies.

# Success Criteria Review

Planned Success Criteria vs Observed Results

Predictive capacity:

- The working plan targeted Ridge R2 > 0.5 and boosted-tree R2 > 0.65. Observed best held-out R2 values are VV=0.902, VH=0.943, and VV-VH=0.707.

- The RMSE < 3 dB target is satisfied comfortably for the completed Sentinel-1 targets: VV=1.260, VH=1.175, VV-VH=1.139.

Cross-modal similarity:

- The plan's optimistic k-NN overlap target (>40%) is not met in pooled form; observed mean overlap at k=10 is 0.032.

- The plan's distance-correlation target (>0.3) is approximately met overall with Pearson r=0.195, and is exceeded within some regions.

Failure analysis:

- The plan expected clear land-use-specific error differences. The completed Phase 3 outputs do show materially different MAE by land-use class.

- Strong global evidence of residual spatial clustering was not established; any isolated Moran's I hits should be treated as exploratory.

# Best Phase 2 Models by Target

| target | feature_set | r2 | rmse | mae | pearson_r |
|---|---|---|---|---|---|
| S1_VH | Embeddings + region + Dynamic World | 0.943 | 1.175 | 0.858 | 0.971 |
| S1_VV | Embeddings + region + Dynamic World | 0.902 | 1.260 | 0.824 | 0.951 |
| S1_VV_div_VH | Embeddings only | 0.707 | 1.139 | 0.796 | 0.841 |

These are the best held-out full-dataset models selected for the project summary.

# Polarization Difference Comparison

| feature_set | model_label | target | training_target | r2 | rmse | mae | pearson_r |
|---|---|---|---|---|---|---|---|
| Embeddings only | Ridge (VH) [Embeddings only] | S1_VH | S1_VH | 0.929 | 1.317 | 0.945 | 0.964 |
| Embeddings only | Ridge (VV) [Embeddings only] | S1_VV | S1_VV | 0.868 | 1.456 | 0.938 | 0.933 |
| Embeddings only | Ridge (VV-VH) [Embeddings only] | S1_VV_div_VH | S1_VV_div_VH | 0.708 | 1.137 | 0.801 | 0.842 |
| Embeddings only | Structural baseline (VV_hat - VH_hat) [Embeddings only] | S1_VV_div_VH | S1_VV+S1_VH | 0.708 | 1.137 | 0.801 | 0.842 |
| Embeddings only | LightGBM (VV-VH) [Embeddings only] | S1_VV_div_VH | S1_VV_div_VH | 0.707 | 1.139 | 0.796 | 0.841 |
| Embeddings + region + Dynamic World | Ridge (VH) [Embeddings + region + Dynamic World] | S1_VH | S1_VH | 0.929 | 1.315 | 0.944 | 0.964 |
| Embeddings + region + Dynamic World | Ridge (VV) [Embeddings + region + Dynamic World] | S1_VV | S1_VV | 0.870 | 1.445 | 0.934 | 0.934 |
| Embeddings + region + Dynamic World | Ridge (VV-VH) [Embeddings + region + Dynamic World] | S1_VV_div_VH | S1_VV_div_VH | 0.708 | 1.137 | 0.801 | 0.842 |
| Embeddings + region + Dynamic World | Structural baseline (VV_hat - VH_hat) [Embeddings + region + Dynamic World] | S1_VV_div_VH | S1_VV+S1_VH | 0.708 | 1.137 | 0.801 | 0.842 |
| Embeddings + region + Dynamic World | LightGBM (VV-VH) [Embeddings + region + Dynamic World] | S1_VV_div_VH | S1_VV_div_VH | 0.699 | 1.154 | 0.798 | 0.836 |

This table compares direct VV-VH prediction, the structural VV_hat - VH_hat baseline, and the reused LightGBM VV-VH result.

# Top S1_VV Failure Modes

| dw_label_name | mae | rmse | bias | p90_abs_error | n |
|---|---|---|---|---|---|
| built | 1.227 | 2.070 | -0.333 | 2.176 | 96 |
| water | 1.011 | 1.500 | 0.020 | 1.957 | 96 |
| shrub_and_scrub | 0.902 | 1.366 | -0.054 | 1.742 | 96 |
| bare | 0.837 | 1.202 | 0.289 | 1.697 | 96 |
| crops | 0.772 | 1.006 | 0.056 | 1.594 | 96 |

The cumulative report focuses on the highest-error S1_VV land-use classes.

# S1_VV Spatial Residual Structure

| region | morans_i | p_value | significant_at_0_05 | n |
|---|---|---|---|---|
| amazon_forest | -0.026 | 0.506 | False | 216 |
| california_coast | 0.092 | 0.026 | True | 216 |
| iowa_ag | 0.012 | 0.742 | False | 216 |
| sf_bay_urban | 0.032 | 0.384 | False | 216 |

Significant spatial clustering indicates geographically structured failure modes.

# Phase 4 Cross-modal Similarity

| k | n_queries | mean_overlap | median_overlap | p10_overlap | p90_overlap |
|---|---|---|---|---|---|
| 5 | 2880 | 0.020 | 0.000 | 0.000 | 0.000 |
| 10 | 2880 | 0.032 | 0.000 | 0.000 | 0.100 |
| 20 | 2880 | 0.053 | 0.000 | 0.000 | 0.150 |

Cross-modal similarity is summarized by k-NN overlap between embedding space and SAR space.

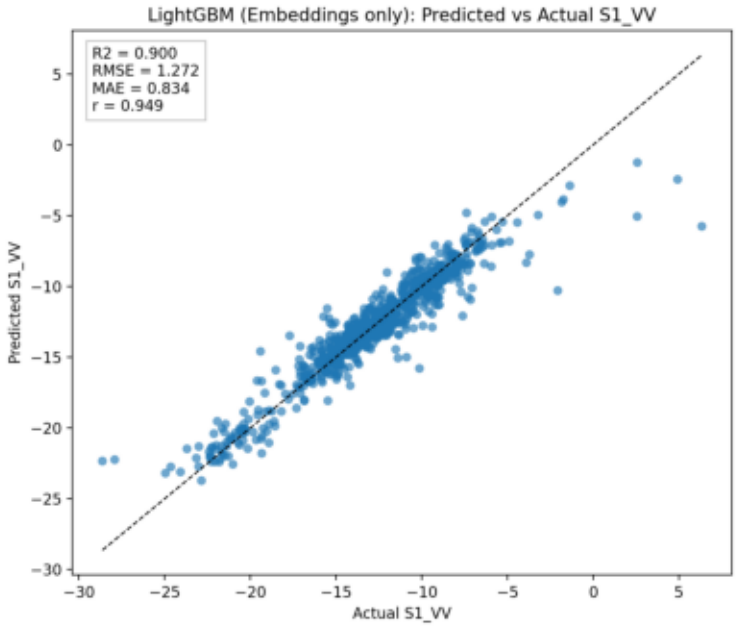
# Top Phase 4 Land-use Alignment at k=10

| dw_label_name | mean_overlap | median_overlap | p10_overlap | p90_overlap | n_queries |
|---|---|---|---|---|---|
| water | 0.098 | 0.100 | 0.000 | 0.300 | 320 |
| snow_and_ice | 0.043 | 0.000 | 0.000 | 0.100 | 320 |
| flooded_vegetation | 0.036 | 0.000 | 0.000 | 0.100 | 320 |
| trees | 0.024 | 0.000 | 0.000 | 0.100 | 320 |
| bare | 0.022 | 0.000 | 0.000 | 0.100 | 320 |

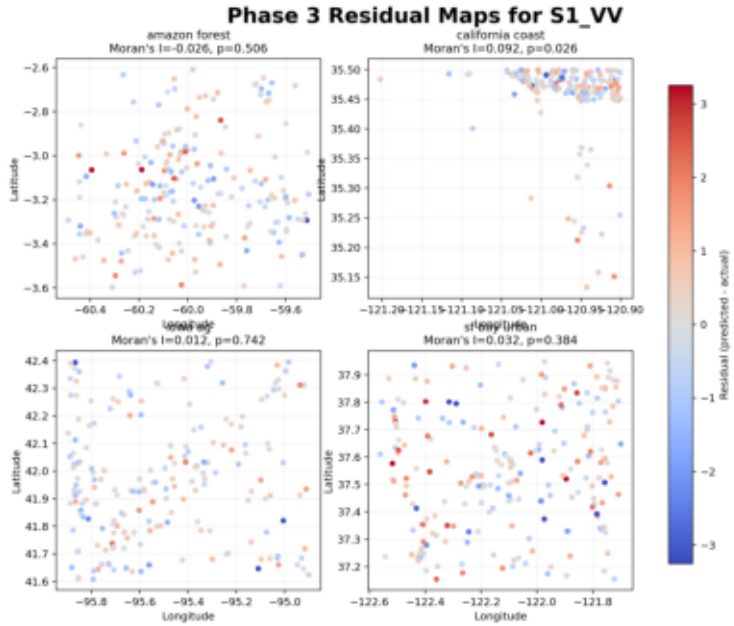
This table highlights which land-use classes show the strongest cross-modal neighbor agreement.

# Project Visual Summary

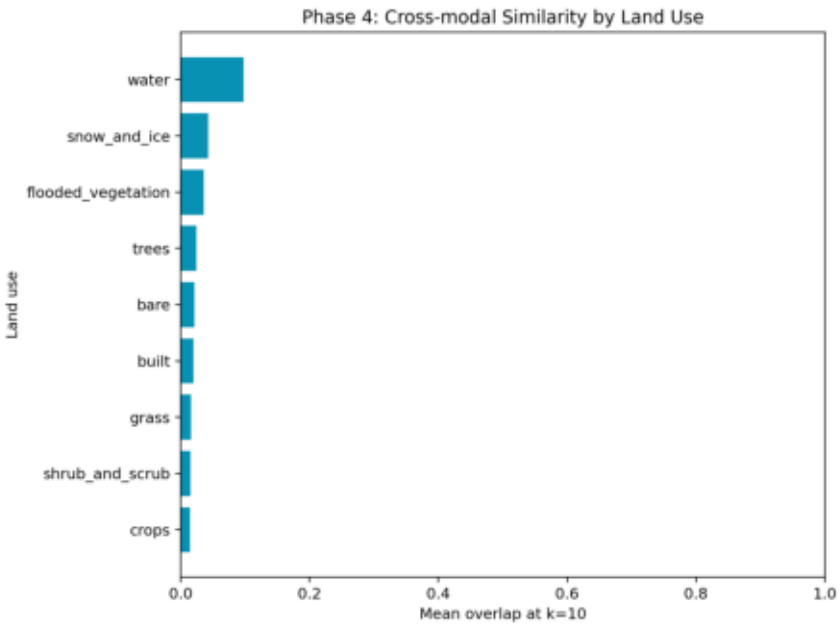
predicted vs actual embedding only S1 VV



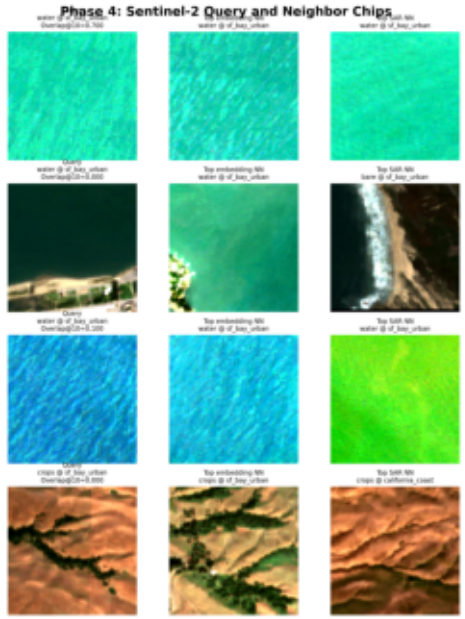

phase3 residual maps S1 VV




phase4 knn overlap by land use k10




phase4 sentinel2 neighbor chips



# Next Steps

Recommended Next Steps

- Add Umbra SAR once co-registered scenes are available so the resolution-sensitivity question from the working plan can be answered directly.

- If the next goal is interpretability, prioritize stable Ridge-style analyses for polarization-difference targets and regional subgroup reporting.

- If the next goal is pure predictive performance, focus on stronger spatial holdouts and additional context features rather than more within-region tuning.

- If the next goal is cross-modal representation learning, use the Phase 4 within-region signal as motivation for joint optical/SAR embedding work.